MOSFET Amplifier Configurations

Unit 5: Field-Effect Transistors

Maxx Seminario

University of Nebraska-Lincoln

Spring 2026

Lecture Overview

Review from Last Lecture

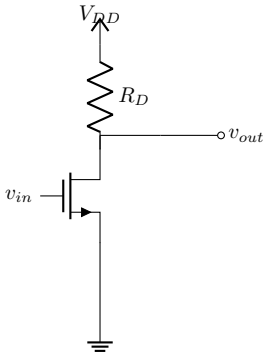
- MOSFET small-signal models
- Transconductance (g_m)
- Output resistance (r_o)
- AC analysis techniques

Today's Topics

- Common-Source (CS) Amplifier
- Common-Drain (CD) Amplifier
- Common-Gate (CG) Amplifier
- Comparison of configurations

Common-Source (CS) Amplifier

Basic Configuration



Key Characteristics

- Input at gate, output at drain
- Source is AC ground (common)
- Most commonly used configuration
- Provides voltage gain with inversion

Basic Formulas

$$A_v \approx -g_m R_D$$
$$R_{in} \approx \infty$$
$$R_{out} \approx R_D$$

Common-Source Amplifier: Voltage Gain

Voltage Gain Derivation

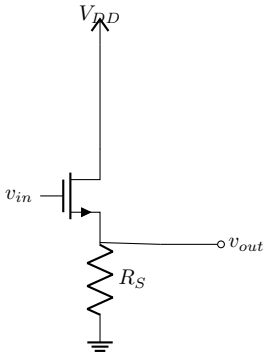
- Output voltage: $v_{out} = -g_m v_{gs} \cdot R_D$
- Input voltage: $v_{in} = v_{gs}$ (infinite gate impedance)
- Voltage gain: $A_v = \frac{v_{out}}{v_{in}} = -g_m R_D$

Key Points

- ☹Negative sign indicates 180° phase inversion
- ☺Gain magnitude can be large
- ☺High input impedance (gate draws no current)
- 😐Moderate output impedance

Common-Drain (CD) Amplifier - Source Follower

Basic Configuration



Key Characteristics

- Input at gate, output at source
- Drain is AC ground (common)
- Also called "source follower"
- Output follows input (no inversion)

Basic Formulas

$$A_v \approx \frac{g_m R_S}{1 + g_m R_S} < 1$$

$$R_{in} \approx \infty$$

$$R_{out} \approx \frac{1}{g_m}$$

Common-Drain Amplifier: Voltage Gain

Voltage Gain Derivation

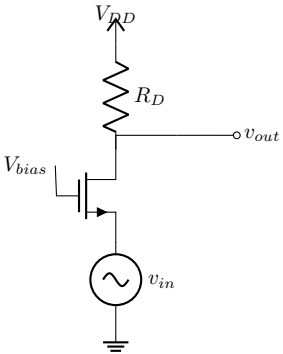
- Output voltage: $v_{out} = g_m v_{gs} \cdot R_S$
- But $v_{gs} = v_{in} - v_{out}$ (negative feedback)
- Solving: $A_v = \frac{v_{out}}{v_{in}} = \frac{g_m R_S}{1+g_m R_S}$

Key Points

- ☺No phase inversion (output in phase with input)
- ☹Voltage gain always less than 1
- ☺Very high input impedance
- ☺Very low output impedance ($\approx 1/g_m$, tens of Ω)
- ☺Excellent buffer for driving low-impedance loads

Common-Gate (CG) Amplifier

Basic Configuration



Key Characteristics

- Input at source, output at drain
- Gate is AC ground (common)
- Less commonly used
- No phase inversion

Basic Formulas

$$A_v \approx g_m R_D$$

$$R_{in} \approx \frac{1}{g_m}$$

$$R_{out} \approx R_D$$

Common-Gate Amplifier: Characteristics

Voltage Gain Derivation

- Output voltage: $v_{out} = g_m v_{gs} \cdot R_D$
- Input voltage: $v_{in} = -v_{gs}$ (gate at AC ground)
- Voltage gain: $A_v = \frac{v_{out}}{v_{in}} = g_m R_D$

Key Points

- ☺No phase inversion (positive gain)
- ☺Voltage gain similar to CS amplifier
- ☹Low input impedance ($\approx 1/g_m$, tens of Ω)
- 😐Moderate output impedance (R_D)

MOSFET Amplifier Configurations: Summary Comparison

Parameter	Common-Source	Common-Drain	Common-Gate
Input Terminal	Gate	Gate	Source
Output Terminal	Drain	Source	Drain
Common Terminal	Source	Drain	Gate
Voltage Gain A_v	$-g_m R_D$ (10-100)	$\frac{g_m R_S}{1 + g_m R_S} < 1$ (0.8-0.99)	$g_m R_D$ (10-100)
Phase Shift	180° (inverted)	0° (in phase)	0° (in phase)
Input Impedance R_{in}	Very High $\approx \infty$	Very High $\approx \infty$	Low $\approx 1/g_m$
Output Impedance R_{out}	Moderate $R_D \| r_o$	Very Low $\frac{1}{g_m} \| R_S \| r_o$	Moderate $R_D \| r_o$
Primary Application	Voltage amplifier	Buffer/Driver	Current buffer

Choosing the Right Configuration

Common-Source (CS)

- Use when: High voltage gain needed, phase inversion acceptable
- Advantages: Large gain, high input impedance
- Disadvantages: Phase inversion, moderate output impedance

Common-Drain (CD) - Source Follower

- Use when: Need to drive low-impedance loads, buffering required
- Advantages: Unity gain, very low output impedance, no inversion
- Disadvantages: Gain less than 1, no voltage amplification

Common-Gate (CG)

- Use when: Wideband operation needed, current buffering
- Advantages: Good gain, no inversion, low Miller capacitance
- Disadvantages: Low input impedance (hard to drive)

Multi-Stage Amplifiers

Cascading Configurations

- Often multiple stages are cascaded to achieve desired performance
- Example: CS followed by CD
  - CS stage provides voltage gain
  - CD stage provides buffering and low output impedance
- Total voltage gain: $A_{v,total} = A_{v,CS} \times A_{v,CD}$
- Design considerations:
  - Input impedance determined by first stage
  - Output impedance determined by last stage
  - Overall gain is product of individual gains

Key Takeaways

1. **Three basic MOSFET amplifier configurations:**
   - Common-Source: High gain, inverted output
   - Common-Drain: Unity gain buffer, very low R_{out}
   - Common-Gate: High gain, low R_{in}, no inversion
2. **Key trade-offs:**
   - Voltage gain vs. input/output impedance
   - Phase relationship (inverted vs. non-inverted)
   - Application-specific requirements
3. **Configuration selection based on:**
   - Required voltage gain
   - Input/output impedance matching
   - Load driving capability
   - Frequency response requirements

Small-Signal Analysis: CS Amplifier (continued)

Derivation Steps:

1. **Identify small-signal voltage**: $v_{gs} = v_{in}$
2. **Apply KCL at drain node**:

$$\frac{v_{out}}{R_D} + \frac{v_{out}}{r_o} + g_m v_{gs} = 0$$

$$v_{out}\left(\frac{1}{R_D} + \frac{1}{r_o}\right) = -g_m v_{in}$$

$$v_{out} = -g_m v_{in}\left(\frac{R_D \cdot r_o}{R_D + r_o}\right)$$

$$v_{out} = -g_m v_{in}(R_D \| r_o)$$

3. **Voltage gain**:

$$A_v = \frac{v_{out}}{v_{in}} = -g_m(R_D \| r_o) \approx -g_m R_D \quad \text{(when } r_o \gg R_D\text{)}$$

Small-Signal Analysis: CD Amplifier (continued)

Derivation Steps:

1. **Identify voltages**: $v_{gs} = v_{in} - v_{out}$ (gate relative to source)
2. **Apply KCL at source node**:

$$g_m v_{gs} + \frac{0 - v_{out}}{r_o} = \frac{v_{out}}{R_S}$$

$$g_m(v_{in} - v_{out}) - \frac{v_{out}}{r_o} = \frac{v_{out}}{R_S}$$

$$g_m v_{in} = v_{out}\left(g_m + \frac{1}{r_o} + \frac{1}{R_S}\right)$$

$$g_m v_{in} = v_{out}\left(g_m + \frac{1}{r_o \| R_S}\right)$$

3. **Voltage gain**:

$$A_v = \frac{v_{out}}{v_{in}} = \frac{g_m}{g_m + \frac{1}{r_o \| R_S}} = \frac{g_m(r_o \| R_S)}{1 + g_m(r_o \| R_S)} \approx \frac{g_m R_S}{1 + g_m R_S} < 1$$

Small-Signal Analysis: CG Amplifier (continued)

Derivation Steps:

1. **Identify small-signal voltage**: $v_{gs} = 0 - v_{in} = -v_{in}$ (gate is at AC ground)
2. **Apply KCL at drain node**:

$$\frac{v_{out}}{R_D} + \frac{v_{out}}{r_o} + g_m v_{gs} = 0$$

$$v_{out}\left(\frac{1}{R_D} + \frac{1}{r_o}\right) = -g_m(-v_{in})$$

$$v_{out}\left(\frac{1}{R_D} + \frac{1}{r_o}\right) = g_m v_{in}$$

$$v_{out} = g_m v_{in}(R_D \| r_o)$$

3. **Voltage gain**:

$$A_v = \frac{v_{out}}{v_{in}} = g_m(R_D \| r_o) \approx g_m R_D \quad \text{(when } r_o \gg R_D\text{)}$$